02. Network Traffic Analysis

1. What Is Network Traffic Analysis?

Definition: Network Traffic Analysis is the process of monitoring and examining data packets travelling across a network to understand communication and identify suspicious activity.

Key Points

- Helps identify suspicious network connections.
- Shows which devices communicate with each other.
- Helps detect malware communication, scanning and data theft.
- Supports troubleshooting and incident response.
- Common tools include Wireshark, tcpdump and Zeek.

Example: A SOC analyst uses Wireshark to investigate why a computer is repeatedly contacting an unfamiliar external IP address.

2. Important Network Protocols

A. DNS (Domain Name System)

Definition: DNS translates domain names into IP addresses.

Example: example.com is resolved to an IP address so a computer can connect to the server.

- Common ports: UDP 53 and TCP 53.
- DNS is used when devices need to resolve domain names.
- DNS can also use encrypted protocols such as DNS over HTTPS (DoH).

B. HTTP (Hypertext Transfer Protocol)

Definition: HTTP is an application-layer protocol used to transfer web pages and other web resources.

- Common port: TCP 80.
- Common methods: GET, POST, PUT and DELETE.
- HTTP traffic is generally unencrypted.
- Analysts may inspect requests, URLs, headers and response codes.

Example: A browser sends an HTTP GET request to retrieve a web page.

C. HTTPS/TLS

Definition: HTTPS is HTTP protected by TLS encryption. TLS helps provide confidentiality, integrity and server authentication.

- Common port: TCP 443.
- TLS protects application data while it travels across the network.
- Analysts may observe TLS handshakes, certificates, server names when exposed, and connection metadata.
- Encrypted application content normally cannot be read directly without authorized decryption.

Example: An online banking website uses HTTPS to protect data exchanged with the browser.

D. ARP (Address Resolution Protocol)

Definition: ARP maps an IPv4 address to a MAC address on a local network.

- ARP operates within the local network.
- An ARP request asks which device owns a particular IPv4 address.
- An ARP reply provides the corresponding MAC address.

Example: A computer uses ARP to find the MAC address of its local gateway.

E. ICMP (Internet Control Message Protocol)

Definition: ICMP is used for network error reporting and diagnostics.

- Used by tools such as ping and traceroute.
- ICMP Echo Request is commonly used to test reachability.
- ICMP Echo Reply indicates that a response was received.
- ICMP traffic does not use TCP or UDP port numbers.

Example: A technician uses ping to check whether another device is reachable.

F. SMB (Server Message Block)

Definition: SMB is a protocol used for file sharing, printer sharing and other network services.

- Common port: TCP 445.
- Often used in Windows networks.
- Provides access to shared folders and network resources.
- Unexpected SMB connections can be relevant to investigations of lateral movement.

Example: An employee accesses a shared folder on a Windows file server.

3. Normal vs Suspicious Network Behaviour

Protocol	Normal Behaviour	Suspicious Behaviour
DNS	Resolving known business or website domains	Repeated requests to unusual domains, unusually long random-looking names, or high volumes of failures
HTTP	Normal GET/POST requests to expected web servers	Requests to suspicious URLs, unexpected uploads, or unusual methods
HTTPS/TLS	Encrypted connections to expected services	Unusual external destinations, unexpected certificate issues, or abnormal connection patterns
ARP	Requests and replies for devices on the local network	Conflicting IP-to-MAC mappings or unexpected ARP replies
ICMP	Occasional ping or diagnostic traffic	Unusual high-volume ICMP traffic or unexpected repeated probing
SMB	Access to authorized file shares	Unexpected SMB access, repeated authentication failures, or unusual connections between endpoints

Important: Suspicious behaviour is an investigation clue, not proof of an attack. Compare it with the network's normal baseline and other available evidence.

4. Useful Wireshark Display Filters

Filter	Purpose
dns	Shows DNS traffic
http	Shows HTTP traffic Wireshark recognizes
tls	Shows TLS traffic
arp	Shows ARP packets
icmp	Shows ICMP traffic
smb	Shows SMB traffic
tcp.port == 445	Shows TCP traffic using port 445
udp.port == 53	Shows UDP traffic using port 53
ip.addr == 192.168.1.10	Shows IPv4 traffic to or from this address
tcp.flags.reset == 1	Shows TCP packets with the RST flag set

Replace 192.168.1.10 with the IP address you are investigating.

5. How to Analyse Traffic in Wireshark

1. Open Wireshark.
2. Select the correct network interface.
3. Start capturing packets.
4. Generate authorized test traffic using ping, nslookup or curl.
5. Enter a display filter such as dns, icmp or tcp.
6. Select a packet to inspect its protocol details.
7. Record the source IP, destination IP, ports, protocol and timestamp.
8. Compare the observed activity with expected behaviour.
9. Save the capture as a .pcapng file for further analysis.

6. Example Investigation

Scenario: A computer sends many DNS requests to an unfamiliar domain.

An analyst should:

- Identify the source device and destination domain.
- Check the request frequency and response codes.
- Examine whether the domain is expected in the environment.
- Review related connections and endpoint logs.
- Document the evidence and escalate according to SOC procedures if necessary.

7. Conclusion

Network Traffic Analysis helps security analysts understand communication between devices and identify unusual patterns. DNS, HTTP, TLS, ARP, ICMP and SMB provide different types of evidence. Wireshark helps inspect these protocols, while comparison with normal network behaviour helps analysts decide what requires further investigation.